

9.5 Kernel Compilation

The title of this section may pop your eyes out! This is one of the best offers that Linux has for its users. Most newbies would wonder if this is necessary. Well, some Linux distributions have kernels specifically designed to run better on Pentium machines.

So, if you have any other processor, and you want to derive better performance out of your box – rebuild the kernel. Kernel compilation is required essentially when certain hardware support is absent; for instance, kernel 2.4.x did not support NTFS partitions by default, but it had the support to read it if enabled. Under such circumstances wherein your PC's peripheral devices are upgraded and a support from the kernel is lacking; rebuilding the same kernel with the support or compiling a new kernel becomes necessary. Compiling a new kernel imparts hardware supports available with the new flavours of Linux and you need not go about installing the entire OS. Note that the looks of the GUI and other software will not change. It is just the kernel that undergoes the changes.

Another point that we want to make here is that Linux is actually the kernel combined with certain utilities to become the OS made by Linus Torvald. The basic Linux OS plus the numerous applications from GNU and Opensource put together, constitutes the Linux Distributions such as Redhat-Fedora, SUSE, Gentoo, etc.

Kernel Compilation or Rebuilding the existing kernel is a huge process. We shall summarise it in point-to-point method and refrain from dragging the topic too far. You will get an idea of the subject and you will also be in a position to research more on the same. In our example, we have used Redhat as our reference distribution.

Necessary files

Kernel sources: Most installations maintain a copy of the source of the kernel in `/usr/src` folder. A new kernel will be available at www.kernel.org.

Listed below are the essential utilities and libraries required for